Assignment 6

Q1

[0.8 point]
Convert the following two grammars into equivalent PDAs using the procedure given at lecture

a)

$$
\begin{aligned}
E &\to E + T \mid T \\
T &\to T \times F \mid F \\
F &\to (E) \mid \mathtt{a}
\end{aligned}
$$

b)

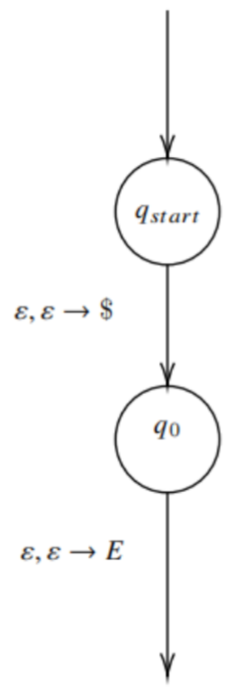
$$
\begin{aligned}
R &\to XRX \mid S \\
S &\to \mathtt{a}T\mathtt{b} \mid \mathtt{b}T\mathtt{a} \\
T &\to XTX \mid X \mid \varepsilon \\
X &\to \mathtt{a} \mid \mathtt{b}
\end{aligned}
$$

A1

a)

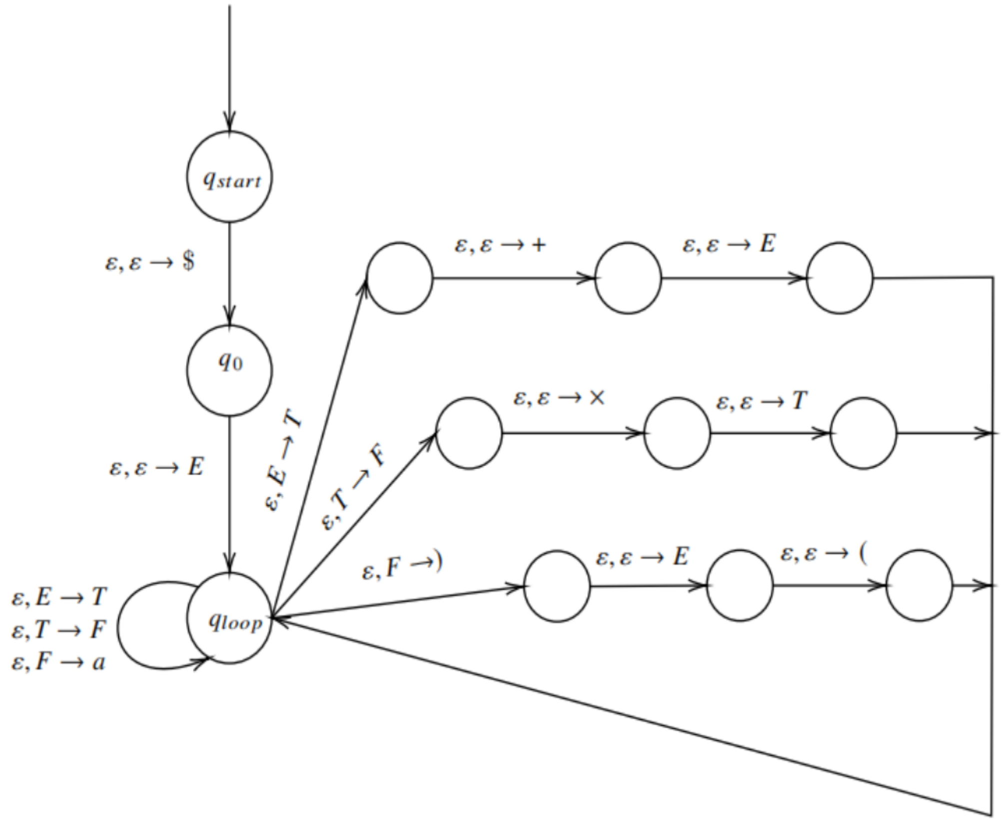
$$
\begin{aligned}
E &\to E + T \mid T \\
T &\to T \times F \mid F \\
F &\to (E) \mid \mathtt{a}
\end{aligned}
$$

First, we push \$ and the start variable E onto the stack.

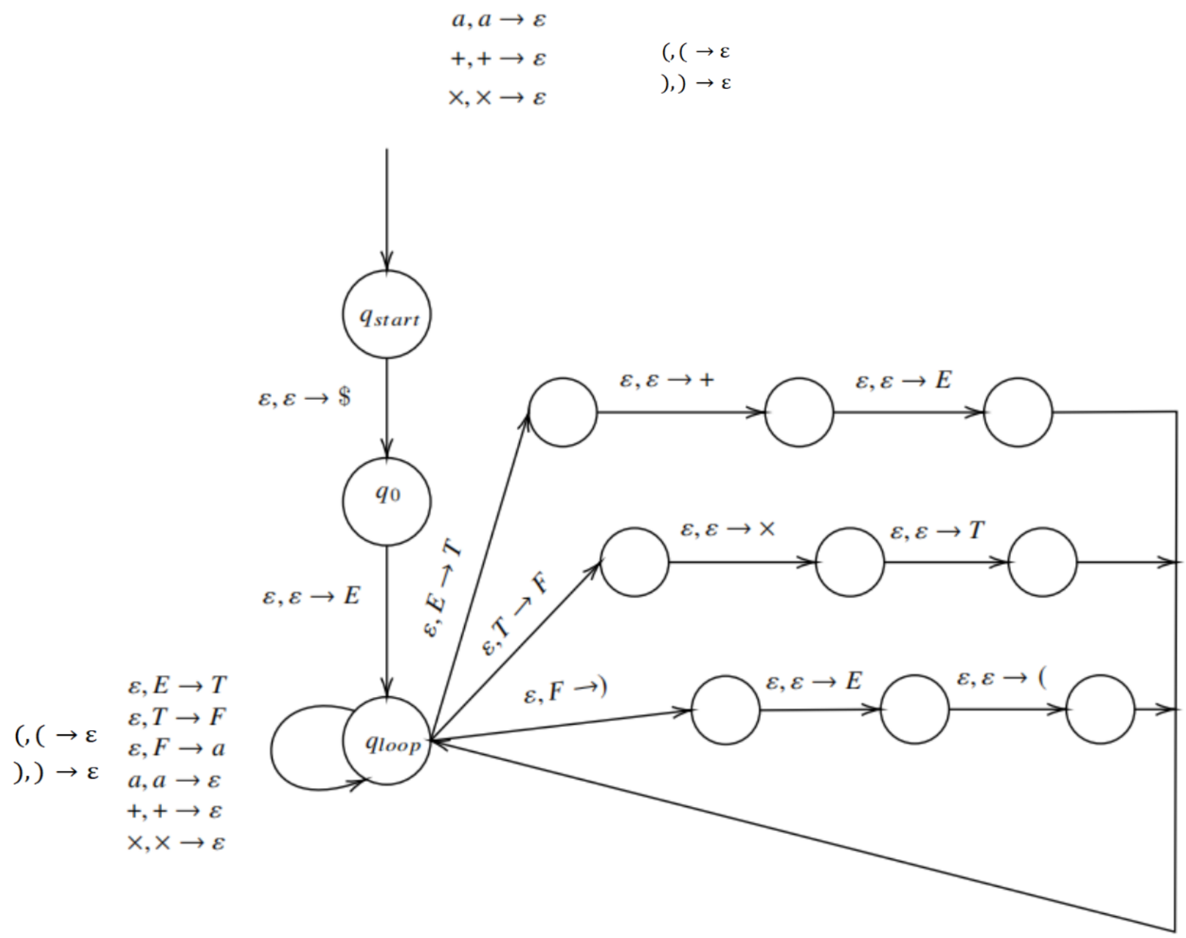


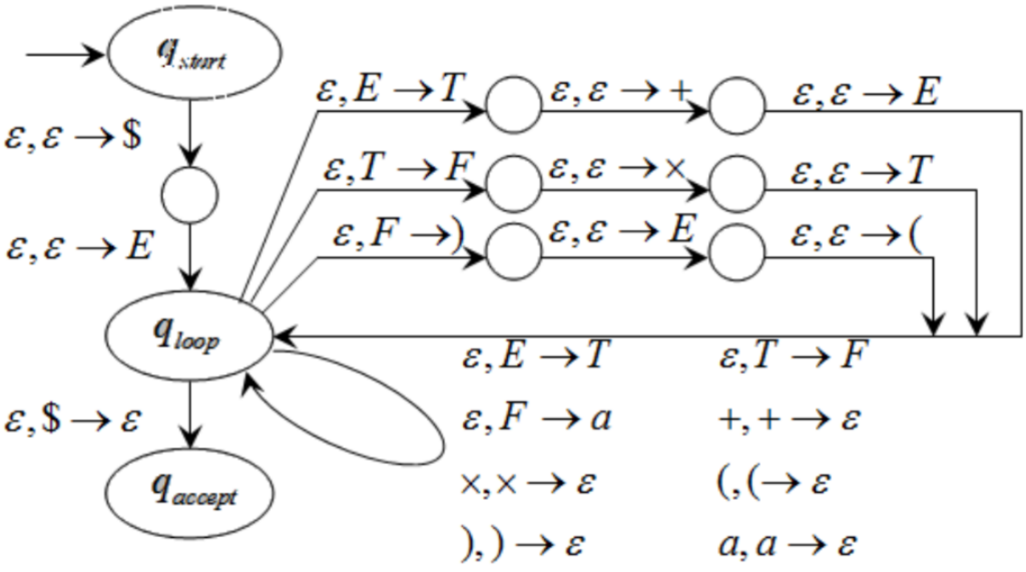
Now we convert the rules of the language.
We handle all the situations where pop is a variable, which means we should add the following path to the diagram.

$$\varepsilon, E \to T,\ \varepsilon, \varepsilon \to +,\ \varepsilon, \varepsilon \to E$$
$$\varepsilon, E \to T$$
$$\varepsilon, T \to F,\ \varepsilon, \varepsilon \to \times,\ \varepsilon, \varepsilon \to T$$
$$\varepsilon, T \to F$$
$$\varepsilon, F \to),\ \varepsilon, \varepsilon \to E,\ \varepsilon, \varepsilon \to ($$
$$\varepsilon, F \to a$$



Then we handle the situation where pop is a terminal, which means we should add the following path to the diagram.





b)

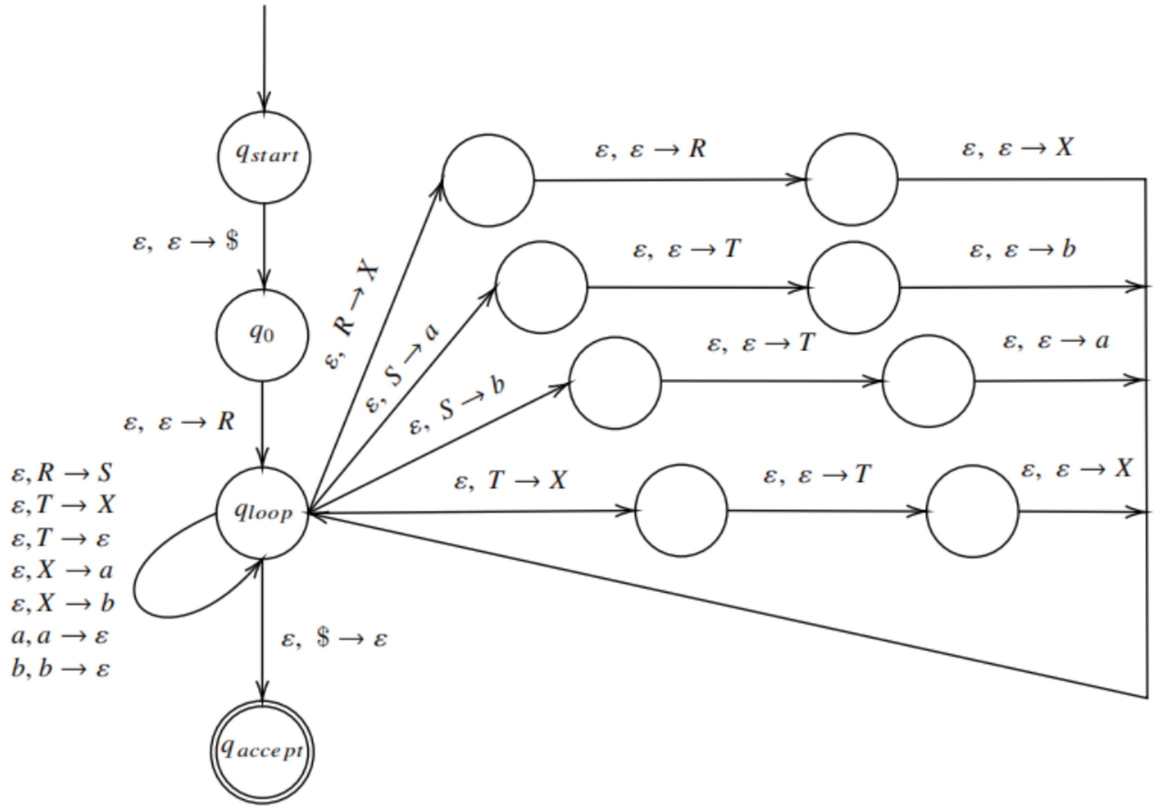
$$
\begin{aligned}
R &\to XRX \mid S \\
S &\to \mathtt{a}T\mathtt{b} \mid \mathtt{b}T\mathtt{a} \\
T &\to XTX \mid X \mid \varepsilon \\
X &\to \mathtt{a} \mid \mathtt{b}
\end{aligned}
$$

We have gone through the approach of converting CFG to PDA in detail at the last question. The procedure here is still the same. To keep the sheet clean and not introduce unnecessary graphs, which can obviously be simplified, I just give the result of the conversion here.

The PDA is

$(\{q_{start}, q_0, q_{loop}, q_{accept}\}, \{a, b\}, \{R, X, S, T, a, b, \$\}, \delta, q_{start}, \{q_{accept}\})$

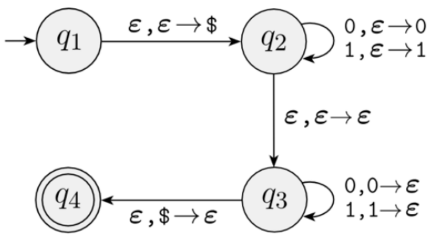
, where δ is in the following form.



Q2

2) [0.7 points]

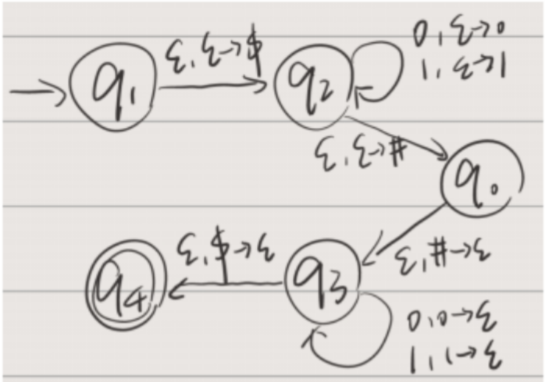
Recall the following PDA for $B = \{ww^{\mathcal{R}} \mid w \in \{0,1\}^*\}$



Convert it into a CFG that recognises B using the proof construction given at lecture.

A2

- Simplify PDA:



- Rule $A_{pq} \to aA_{rs}b$:

  $A_{q_1 q_4} \to \varepsilon A_{q_2 q_3} \varepsilon$

  $A_{q_2 q_3} \to 0A_{q_2 q_3}0 \mid 1A_{q_2 q_3}1 \mid \varepsilon A_{q_0 q_0} \varepsilon$

- Rule $A_{pq} \to A_{pr}A_{rq}$:

  $A_{q_0 q_0} \to A_{q_0 q_0}A_{q_0 q_0} \mid A_{q_0 q_1}A_{q_1 q_0} \mid A_{q_0 q_2}A_{q_2 q_0} \mid A_{q_0 q_3}A_{q_3 q_0} \mid A_{q_0 q_4}A_{q_4 q_0}$

  $A_{q_0 q_1} \to A_{q_0 q_0}A_{q_0 q_1} \mid A_{q_0 q_1}A_{q_1 q_1} \mid A_{q_0 q_2}A_{q_2 q_1} \mid A_{q_0 q_3}A_{q_3 q_1} \mid A_{q_0 q_4}A_{q_4 q_1}$

  $A_{q_0 q_2} \to A_{q_0 q_0}A_{q_0 q_2} \mid A_{q_0 q_1}A_{q_1 q_2} \mid A_{q_0 q_2}A_{q_2 q_2} \mid A_{q_0 q_3}A_{q_3 q_2} \mid A_{q_0 q_4}A_{q_4 q_2}$

  ......

  $A_{q_4 q_2} \to A_{q_4 q_0}A_{q_0 q_2} \mid A_{q_4 q_1}A_{q_1 q_2} \mid A_{q_4 q_2}A_{q_2 q_2} \mid A_{q_4 q_3}A_{q_3 q_2} \mid A_{q_4 q_4}A_{q_4 q_2}$

  $A_{q_4 q_3} \to A_{q_4 q_0}A_{q_0 q_3} \mid A_{q_4 q_1}A_{q_1 q_3} \mid A_{q_4 q_2}A_{q_2 q_3} \mid A_{q_4 q_3}A_{q_3 q_3} \mid A_{q_4 q_4}A_{q_4 q_3}$

  $A_{q_4 q_4} \to A_{q_4 q_0}A_{q_0 q_4} \mid A_{q_4 q_1}A_{q_1 q_4} \mid A_{q_4 q_2}A_{q_2 q_4} \mid A_{q_4 q_3}A_{q_3 q_4} \mid A_{q_4 q_4}A_{q_4 q_4}$

- Rule $A_{pp} \to \varepsilon$:

  $A_{q_0 q_0} \to \varepsilon \quad A_{q_1 q_1} \to \varepsilon \quad A_{q_2 q_2} \to \varepsilon \quad A_{q_3 q_3} \to \varepsilon \quad A_{q_4 q_4} \to \varepsilon$

Q3

3) [0.3 points]
   Show that the class of context free languages is closed under the regular operations, union, concatenation and star.

A3

Prove by CFG:

By definition, A is a Context Free Language if $A = L(G)$ for some CFG G.

Suppose A and B are Context Free Languages where $A = L(G_1)$ for CFG G_1 and $B = L(G_2)$ for CFG G_2.

Let $G_1 = (V_1, \Sigma_1, R_1, S_1)$ and $G_2 = (V_2, \Sigma_2, R_2, S_2)$.

First, we prove the class of context free languages is closed under union.

Suppose C is the union of A and B. Then C is the language for

$$G_3 = (V_1 \cup V_2 \cup \{S\}, \Sigma_1 \cup \Sigma_2, R_1 \cup R_2 \cup \{S \to S_1 \mid S_2\}, S).$$

Obviously, G_3 is a CFG so C is a context free language.

Second, we prove the class of context free languages is closed under concatenation.

Suppose D is the concatenation of A and B. Then D is the language for

$$G_4 = (V_1 \cup V_2 \cup \{S\}, \Sigma_1 \cup \Sigma_2, R_1 \cup R_2 \cup \{S \to S_1 S_2\}, S).$$

Obviously, G_4 is a CFG so D is a context free language.

Finally, we prove the class of context free languages is closed under star.

Suppose E is the star of A. Then E is the language for

$$G_5 = (V_1 \cup \{S\}, \Sigma_1, R_1 \cup \{S \to S_1 S \mid \varepsilon\}, S).$$

Obviously, G_5 is a CFG so E is a context free language.

Therefore, the class of context free languages is closed under union, concatenation and star.

Prove by PDA:

First we find the context free grammar for the corresponding context free language. Then by the construction proof, we convert the context free grammar to PDAs. Next we show the union, concatenation and star of the PDAs. For convention, we modify the PDA to give the following feature: It empties its stack before acception. Assume the result is $A \oplus B$ where $* \in \{\cup, \circ\}$ or A^*, where $A,\ B$ are PDAs.

1. For union, we create a new start state q_0 and add $\delta(q_0, \varepsilon, \varepsilon) = (r, \varepsilon)$ for all $r \in \{\text{The start state of A and B}\}$. The ensures if A or B accepts the string, the new PDA will also accepts.

2. For concatenation, we add $\delta(a, \varepsilon, \varepsilon) = (b, \varepsilon)$ for all $a \in \{\text{The accept states of A}\}$, $b \in \{\text{The start state of B}\}$, and remove all accept state of A.

3. For star, we create a new start state and also the accept state q_0, and add $\delta(q_0, \varepsilon, \varepsilon) = (r, \varepsilon)$ where r is the start state of A. Then we add $\delta(a, \varepsilon, \varepsilon) = (a_0, \varepsilon)$ where $a \in \{\text{The accept states of A}\}$ and $a_0 \in \{\text{The start state of A}\}$.

Because in all cases the stack is empty when accepts, so it can accept a valid string again if it goes back to the start state. Since CFG and PDA are equivalent, and we show that we can construct a new PDA that recognizes $A \cup B,\ A \circ B,\ \text{and}\ A^*$, the context free languages is closed under union, concatenation, and star.

Q4

4) [0.7 points]
Use the results of the previous exercise to give another proof that every regular language is context free by showing how to convert a regular expression directly to an equivalent context free grammar.

A4

By the definition of regular expressions, R is a regular expression if R is:

1. a for some a in the alphabet Σ,
2. ε,
3. Φ,
4. $(R_1 \cup R_2)$, where R_1 and R_2 are regular expressions,
5. $(R_1 \circ R_2)$, where R_1 and R_2 are regular expressions, or
6. (R_1^*), where R_1 is a regular expression.

For case 1, $R = a$ can be generated by a context free grammar $G = (\{S\}, \Sigma, \{S \to a\}, S)$.
For case 2, $R = \varepsilon$ can be generated by a context free grammar $G = (\{S\}, \Sigma, \{S \to \varepsilon\}, S)$.
For case 3, $R = \Phi$ can be generated by a context free grammar $G = (\{S\}, \Sigma, \Phi, S)$.
For case 4, 5 and 6, we have proved context free languages is closed under union, concatenation and star in question 4, and also constructed context free grammar for them.
To sum up, we can convert a regular expression to an equivalent context free grammar for all the cases, so that we proved that every regular language is context free.

Q5

5) [0.3 points] Prove that if a CFG G is in Chomsky Normal Form then every derivation of a string w has $2|w| - 1$ steps.

A context-free grammar is in ***Chomsky normal form*** if every rule is of the form

$$A \to BC$$
$$A \to a$$

where a is any terminal and A, B, and C are any variables—except that B and C may not be the start variable. In addition, we permit the rule $S \to \varepsilon$, where S is the start variable.

A5

It can be proved applying the induction method by on the length of the string w.

Basic case: for $|w| = 1$, consider the string a of length 1. So the derivation for this is $S \to a$, where S is the starting symbol. Hence the derivation of this has $2 * 1 - 1 = 1$ step.

Inductive hypothesis: now suppose that the statement is true for all the strings of length $|w| \leq k$.

Inductive case: now consider $|w| = k + 1$. Suppose $w = xy$ where $|x| + |y| = |w|$ and $|x|, |y| > 0$. Then the derivation of w must look like:

$S \to AB$

$A \overset{*}{\Rightarrow} x$

$B \overset{*}{\Rightarrow} y$

By the inductive hypothesis, since $|x| \leq k$, then the derivation from A to the string x must take $2|x| - 1$ steps. Samely, the derivation from B to y will also take $2|y| - 1$ steps. So the derivation of w must take $1 + 2|x| - 1 + 2|y| - 1 = 2(|x| + |y|) - 1 = 2|w| - 1$ steps.

Q6

6) [0.6 points]

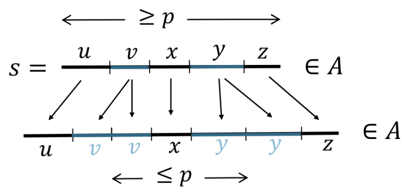
Use the pumping lemma to show that the following languages are not context free.

a. $\{0^n 1^n 0^n 1^n \mid n \geq 0\}$

b. $\{0^n \# 0^{2n} \# 0^{3n} \mid n \geq 0\}$

Pumping Lemma for CFLs: For every CFL A, there is a p such that if $s \in A$ and $|s| \geq p$ then $s = uvxyz$ where

1) $uv^i xy^i z \in A$ for all $i \geq 0$
2) $vy \neq \varepsilon$
3) $|vxy| \leq p$



Informally: All long strings in A are pumpable and stay in A.

- Condition 2 says that either v or y is not the empty string;
- Condition 3 says that vxy has length at most p

A6

a)

Let p be the pumping length of B given by the pumping lemma. To show that B is not a CFL, it is enough to show that a string $s = 0^p 1^p 0^p 1^p$ cannot be pumped. Consider s is of the form $uvxyz$.

- If both v and y contain at most one type of alphabet symbol, then uv^2xy^2z will not be of the form $0^n 1^n 0^n 1^n$.
- If either v or y contains more than one type of alphabet symbol, the string uv^2xy^2z does not contain the symbol in correct order.

Thus this language is not context free.

b)

(b) Let $B = \{0^n \# 0^{2n} \# 0^{3n} \mid n \geq 0\}$. Let p be the pumping length given by the pumping lemma. Let $s = 0^p \# 0^{2p} \# 0^{3p}$. We show that $s = uvxyz$ cannot be pumped.

Neither v nor y can contain #, otherwise uv^2xy^2z contains more than two #s. Therefore, if we divide s into three segments by #'s: 0^p, 0^{2p}, and 0^{3p}, at least one of the segments is not contained within either v or y. Hence uv^2xy^2z is not in B because the $1:2:3$ length ratio of the segments is not maintained.

Q7

7) [0.4 points]

Use the languages $A = \{a^m, b^n, c^n \mid m, n \geq 0\}$ and $B = \{a^n, b^n, c^m \mid m, n \geq 0\}$ to show that the class of context free is not closed under intersection.

A7

Note: the comma "," is a typo. A should be $\{a^m b^n c^n\}$ and B should be $\{a^n b^n c^m\}$. Anyway, it won't affect the idea of the proof.

Proof:

$A \cap B = \{a^k b^k c^k \mid k \geq 0\}$. And this has already been shown in the lecture that is not conext free.

Let $B = \{0^k 1^k 2^k \mid k \geq 0\}$

Show: B is not a CFL

Proof by Contradiction:

Assume (to get a contradiction) that B is a CFL .
The CFL pumping lemma gives p as above. Let $s = 0^p 1^p 2^p \in B$.
Pumping lemma says that can divide $s = uvxyz$ satisfying the 3 conditions.
Condition 3 ($|vxy| \leq p$) implies that vxy cannot contain both 0s and 2s.
So uv^2xy^2z has unequal numbers of 0s, 1s, and 2s.
Thus $uv^2xy^2z \notin B$, violating Condition 1. Contradiction!
Therefore our assumption (B is a CFL) is false. We conclude that B is not a CFL .

$s = 00\cdots0011\cdots1122\cdots22$

$u \mid v \mid x \mid y \mid z$

$\leftarrow \leq p \rightarrow$

Q8

8) [0.4 points]

Use the result of the previous exercise as well as De Morgan's law:

a. For any two sets A and B, $\overline{A \cup B} = \overline{A} \cap \overline{B}$

to show that the class of context-free languages is not closed under complementation.

A8

Assumue context free languages is closed under complementation. Let A, B be two context free languages. Then $\overline{A} \cup \overline{B}$ is context free, because context free languages is closed under union as proved previously. According to De Morgan's Law, $\overline{\overline{A} \cup \overline{B}} = A \cap B$ is context free because of the closure under complementation (assumption). But as shown in `Question 7`, context free languages is not closed under intersection, $A \cap B$ bis not context free as the previous example. Contradict to the assumption. Therefore context free languages is not closed under complementation.

Q9

9) [0.4 points]

Let C be a context-free language and R be a regular language. Prove that the language $C \cap R$ is context free.

A9

(a) Let C be a context-free language and R be a regular language. Let P be the PDA that recognizes C, and D be the DFA that recognizes R. If Q is the set of states of P and Q' is the set of states of D, we construct a PDA P' that recognizes $C \cap R$ with the set of states $Q \times Q'$. P' will do what P does and also keep track of the states of D. It accepts a string w if and only if it stops at a state $q \in F_P \times F_D$, where F_P is the set of accept states of P and F_D is the set of accept states of D. Since $C \cap R$ is recognized by P', it is context free.